Introduction to Probability and Statistics

Exercises Lecture 5

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Exercise 1: Adapted from Chapter 8, problem 25

It is claimed that a certain type of bipolar transistor has a mean value of current gain that is at least 210. A sample of these transistors is tested. If the sample mean value of the current gain is 200 and it is known that the standard deviation is 35, would the claim be rejected at the 5 percent level of significance if

a) The sample size is 25?
b) The sample size is 64?

Tips: First, formulate the null hypothesis H_0. Then, identify the type of test:

- Is it one-side or two-side test?
- What is the test statistic T that you should use?
- Will you compare T to the threshold c or the p-value to α?

Try to interpret the results.

Exercise 2: Chapter 8, problem 56

According to the U.S. Bureau of Census, 25.5 percent of the population of those age 18 or over smoked in 1990. A scientist has recently claimed that this percentage has since increased, and to prove her claim she randomly sampled 500 individuals from this population. If 138 of them were smokers, is her claim proved? Use the 5 percent level of significance.

Tips: Which type of distribution do the RVs have when there are only two options for the outcomes? What is the type of distribution that results from the sum of several of these RVs?

Several of the exercises were taken from the book "Introduction to Probability and Statistics for Engineers and Scientists" by Sheldon M. Ross, 3rd Ed.

Exercise 3: The students from last year were asked to choose between the following two options for the rest of the course using an online poll.

1) Israel gives the rest of the theory on March 26. Students do the hypothesis testing exercises on their own. We do the workshop exercises on April 18.
2) Israel gives half of the theory on March 26 and the rest on April 18. We do the exercises for hypothesis testing and the workshop mixed with the lectures.

Naturally, not all the students responded to the poll. As Fig. 1 with a total of $n = 32$ responses, 15 of them selecting option 2).

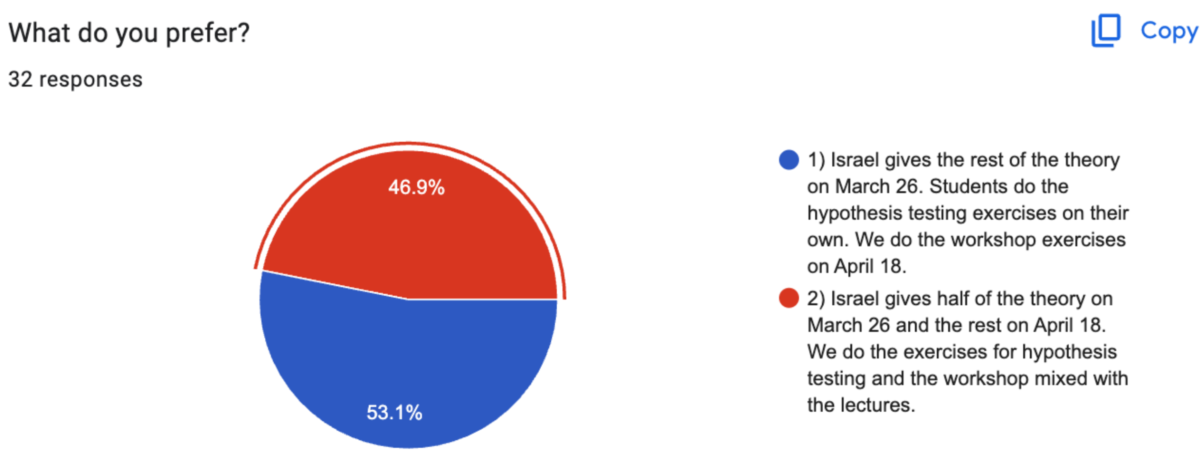

Fig. 1. Results of the voting for the next sessions of statistics.

To make a decision that is fair, Israel decided to conduct a test. First, he defined $X_i \sim \text{Bernoulli}(p)$, where $X_i = 0$ if a student selects 1) and $X_i = 1$ the students selects 2). With this, he set $p_0 = 0.5$, defined the null hypothesis H_0: "the students prefer option 2)", which can be formulated as $p \geq p_0$. Then, he estimated

$$\hat{p}_{32} = \frac{1}{32} \sum_{i=1}^{n} X_i = 0.469$$

a) Perform a one-sided test with $\alpha = 0.05$ using the null hypothesis H_0 and calculating the p-value for the Binomial distribution $v = P(X \leq k; n, p_0)$, where $k = 15$ is the number of students that chose option 2). Can you reject Israel's hypothesis?
b) Change the null hypothesis to be H_0': "the students prefer option 1)". What is the mathematical formulation of this? Can you reject this hypothesis with the collected evidence?

c) Assume that the result of the poll is now $k = 11$, meaning that only eleven students preferred option 2). Can the original null hypothesis H_0 be rejected?